LESSON 5.5

Determine the Volume of Rectangular Prisms

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the volume of a rectangular prism with whole-number edges using length × width × height.

WHY THIS MATTERS Your team is building time capsule containers to store class memories. Each container is a rectangular prism, and you need to figure out how much each one can hold. That means calculating the volume using whole number dimensions! Container A is 5 inches long, 3 inches wide, and 4 inches tall. How much can it hold?

REMEMBER

Determine the Volume of Rectangular Prisms

Volume of Rectangular Prisms (Whole Numbers). Formula: V = Length × Width × Height = Base Area × Height (V = l · w · h = B · h) Volume is how much space is inside a 3D shape. For a rectangular prism (a box), volume = length × width × height, measured in cubic units.

1. Identify the length, width, and height of the box.
2. Multiply base area (l × w) by height (h).
3. Label answer in cubic units (e.g. in³, cm³, ft³)

Words: **Volume *(Volumen)*** · **Rectangular prism *(Prisma rectangular)*** · **Cubic units *(Unidades cúbicas)*** · **Length, width, height *(Largo, ancho, altura)*** · **Net *(Plantilla (desarrollo plano))***

1 **Circle the letter of the best answer. Show how you know.**

A box has a volume of 72 in³. Its base is 6 in by 4 in. What is its height?

(A) 2 in (B) 3 in (C) 12 in (D) 18 in

HINT You know the volume (72 in³) and the base (6 in by 4 in). Only the height is unknown.

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

2 **Circle the letter of the best answer. Show how you know.**

To find the area of a composite figure, you first…

(A) measure around the outside

(B) multiply all the side lengths

(C) find the longest side

(D) break it into basic shapes you know

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

A figure is a 6 by 4 rectangle with a 2 by 3 rectangle attached. What is the total area?

(A) 15 square units

(B) 24 square units

(C) 30 square units

(D) 144 square units

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

4 **Circle the letter of the best answer. Show how you know.**

A 10 by 8 rectangular garden has a 3 by 3 square patio cut out of it. What area is left?

(A) 62 square units

(B) 71 square units

(C) 80 square units

(D) 89 square units

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

5 **Circle the letter of the best answer. Show how you know.**

A shape is made of a 5 by 4 rectangle joined to a triangle with base 5 and height 4. What is the total area?

(A) 20 square units
(B) 25 square units
(C) 30 square units
(D) 40 square units

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

6 **Circle the letter of the best answer. Show how you know.**

A moving company uses rectangular boxes. A large box is 24 inches long, 18 inches wide, and 18 inches tall. A medium box is 18 inches long, 14 inches wide, and 12 inches tall. What is the volume of the large box?

(A) 60 in³
(B) 3,024 in³
(C) 4,752 in³
(D) 7,776 in³

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

EXPLAIN YOUR THINKING

Container C holds the most (96 in³) and Container B holds the least (50 in³). What do you notice about the dimensions that make one volume larger than another?

SENTENCE STARTER Container C has the largest volume because ___ × ___ × ___ = ___. Compared to Container B, the dimensions ___ produce a ___ volume because ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? □ 4 · I can teach it □ 3 · I've got it □ 2 · I need a hint □ 1 · I need help

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LEARNING TARGET I can find the volume of a rectangular prism with whole-number edges using length × width × height.

WHY THIS MATTERS Your team is building time capsule containers to store class memories. Each container is a rectangular prism, and you need to figure out how much each one can hold. That means calculating the volume using whole number dimensions! Container A is 5 inches long, 3 inches wide, and 4 inches tall. How much can it hold?

PART 1 Understand the idea

models and meaning

1 Write the letter of the matching item on each line.

Match each capsule's dimensions to its volume. Use V = length × width × height.

_____ 7 × 2 × 3	(A) 42 cubic units
_____ 6 × 5 × 2	(B) 60 cubic units
_____ 9 × 4 × 2	(C) 72 cubic units
_____ 5 × 3 × 2	(D) 30 cubic units

2 Write the name of the correct group on each line.

A capsule is filled with unit cubes. Sort each statement by whether it describes COUNTING cubes by layers or finding volume with the FORMULA.

Groups: Counting Cubes by Layers | Using the Volume Formula

_____ Count 4 layers of 6 cubes each

_____ Add up the cubes in every layer

_____ Stack layers until the box is full, then count

_____ Use V = length × width × height

_____ Multiply the three edge lengths together

_____ Compute 6 × 5 × 2 = 60 cubic units

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

A moving company uses rectangular boxes. A large box is 24 inches long, 18 inches wide, and 18 inches tall. A medium box is 18 inches long, 14 inches wide, and 12 inches tall. What is the volume of the medium box?

(A) 2,520 in³
(B) 3,024 in³
(C) 3,240 in³
(D) 4,752 in³

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

4 **Circle the letter of the best answer. Show how you know.**

A moving company uses rectangular boxes. A large box is 24 inches long, 18 inches wide, and 18 inches tall. A medium box is 18 inches long, 14 inches wide, and 12 inches tall. How many more cubic inches does the large box hold?

(A) 1,728 in³
(B) 3,024 in³
(C) 4,752 in³
(D) 10,800 in³

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

A time capsule box is 5 units long, 3 units wide, and 2 units tall. How many unit cubes fill it?

(A) 10 cubic units
(B) 15 cubic units
(C) 25 cubic units
(D) 30 cubic units

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

You stack unit cubes in a capsule: each layer has 6 cubes, and there are 4 layers. What is the volume?

(A) 10 cubic units
(B) 12 cubic units
(C) 18 cubic units
(D) 24 cubic units

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

EXPLAIN YOUR THINKING

Container C holds the most (96 in³) and Container B holds the least (50 in³). What do you notice about the dimensions that make one volume larger than another?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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